Day 32 – PM session

Sudhanshu Biswas

Part B: Stretch Problem — Gradient Boosting Preview

Boosting vs Bagging

Bagging (Random Forest) trains B independent trees in parallel on bootstrap samples and aggregates predictions by majority vote or averaging — errors from individual trees are uncorrelated so they cancel out. Boosting trains trees sequentially, where each tree is fit on the residuals (errors) of the ensemble so far. Each new tree specifically targets the mistakes the current model makes; the final prediction is a weighted sum of all trees. Bagging reduces variance; boosting primarily reduces bias, but is more prone to overfitting if not regularised.

Reference

StatQuest with Josh Starmer — "Gradient Boost Part 1: Regression Main Ideas"

https://www.youtube.com/watch?v=3CC4N4z3GJc

Clear visual walkthrough of how residuals drive sequential tree building. Covers regression first, then extends to classification — good pre-read before tomorrow's session.